



Framing Requirements Work as Learning

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WHY IS REQUIREMENTS work not more perceived as learning? Rarely at conferences, seminars, or even informal get-togethers do requirements people frame their work to involve learning. I can think of some reasons for this—we are, after all, an engineering discipline—but still. Which is why it was refreshing to see a requirements person stand up and do just that at a recent conference: Hermann Kaindl, from the Vienna University of Technology.

Well, to be honest, Hermann didn't actually say that we should frame requirements work as a learning process, but from my seat in the back, he might as well have. He actually argued for framing requirements work as knowl-

in the audience looked a little vexed. After all, making the tacit explicit was something that we've been doing since the dawn of our discipline. Not everyone saw this new angle—their loss.

But my concern was different: Why didn't Hermann follow his argument through to its logical conclusion? Why not, if you view requirements work through the lens of knowledge management, view it as a learning process, in that analysts need to learn about the domain, its problems, stakeholders, and their motives? And if organizational learning, whatever that is, is taking place, is individual learning by the analysts not likewise taking place?

individual learning—has Maiden lost the plot?

Most definitely not—I'm being serious.

Our community's perceptions of the social processes in requirements work have evolved considerably over the past 20 years. Our earliest, simplistic view in which an analyst elicits fully formed requirements from willing stakeholders has given way to the realization that we need to acquire requirements from stakeholders through dialogue, observation, and better understanding based on immersion and contextual inquiry. More recently, some of us have argued that requirements also need to be created, invented, or just made up.

All of these social processes involve some degree of learning—about stakeholders, their processes, and their workplace. Learning has become a very big elephant in our requirements room.

I believe that framing requirements work as learning opens up new opportunities. One such opportunity is to apply existing theories and models of learning to our work. Of course, learning has been well researched, and the discerning requirements analyst is spoiled for choice.

Consider reflective learning. It's actually part of reflective practice, which

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edge management.¹ He and his coauthor Lukas Pilat cast requirements engineering as a knowledge management process to adopt the spiral of knowledge and transform it from tacit to explicit. Some of the requirements people

Reflections

At this point, some of you might be checking the strap line at the top of this page and wondering if this is the Requirements column—or if it's even *IEEE Software*. Organizational and

is continuous learning from reflecting on past actions. Some argue that it is a defining characteristic of professional practice, hence we requirements folks should be aware of it. Reflective learning from past actions describes “those intellectual and affective activities in which individuals engage to explore their experiences in order to lead to new understandings and appreciations.”² A learner reevaluates a past experience by attending to its various aspects, thereby producing outcomes.

Figure 1 shows the reflective learning process developed by David Boud and his colleagues.² Experiences refer both to single experiences (of specific events or situations) and general experience in the form of knowledge/skills/attitudes collected and developed over time. It can be seen as a continuous “flow” of which people can be more or less conscious.

Reality Check

So what might the implications of reflective learning be on requirements work? Well, one is that analysts should be encouraged to reflect more on the work undertaken, to support their learning. Most of us are in a rush, chasing another milestone or the next deadline. We have little if any time to look backward. Perhaps we need to encourage more reflective practices in requirements work. If we did, what might these practices look like?

The Boud model in Figure 1 advocates returning to experiences, attending to our feelings about them, and reevaluating them to gain new perspectives and change our behavior. This suggests that learning about requirements work can take place in two related dimensions. One is learning about our own requirements tasks—how to do the work better in the future. The other is learning about the application domain and requirements on future versions of it.

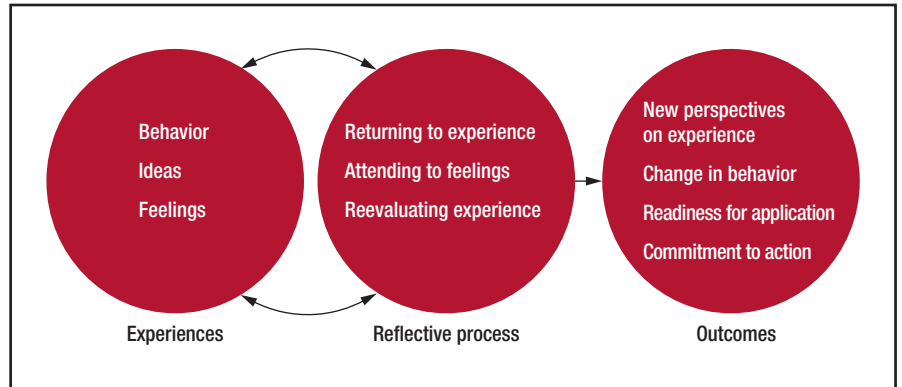


FIGURE 1. The process of reflective learning. Experience can be seen as a continuous “flow” of which people can be more or less conscious.

Let’s focus on that latter point: reflective learning could be applied to our workplace observations. Analysts often document the results of such observations using notes, photographs, and critical-incident reports. The focus is often on being objective, but what if we reflect on the observation experience and attend to our emotions, as we might during an emotional design process? Might we learn something more about the domain?

Likewise, during walkthroughs of scenarios or use cases, analysts generate outcomes, namely, new requirements or emerging inconsistencies or conflicts between requirements. Again, the focus tends to be on objective outcomes. But what if, later on, we return to an experience and replay it in our minds and do the walkthrough again? Might we learn something more general about the future system?

Agile requirements processes could also benefit from more reflective learning. The focus on the next sprint reduces time for learning about the requirements before the code is developed. Is this a good thing? I doubt it—you need some space to reflect on the impact of emerging code on the delivery of important qualities required by the final product.

So, after reading this column, what are your immediate reflections (ha!)? Consider your emotions—anger, delight, frustration—can you reevaluate your reading experience?

Let me guide your learning. Reflection is a characteristic of good professional practice. If, as professionals, we’re expected to do it, then we should embrace it and apply it directly to requirements work. We might learn something. ☞

References

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2. D. Boud, R. Keogh, and D. Walker, *Promoting Reflection Experience into Learning*, Routledge Falmer, 1985, pp. 18–40.

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